

CHAPTER

5

CAPACITY EXPANSION PLANNING ON THE NORTH CAROLINA ENERGY SYSTEM: ANALYZING POLICIES, COST TRAJECTORIES, AND HURRICANE RISKS

This chapter explores the strategic planning of the North Carolina power sector through both deterministic and stochastic capacity expansion models, examining the period from 2023 to 2050. It evaluates the impact of various government policies, alternative technology forecasts from the NREL-ATB, and fuel price trajectories from the EIA-AEO. The risk of hurricane damage to the power system planning is also explored under alternative policy and technology trajectories using stochastic optimization. Our findings suggest that while nuclear and natural gas will remain the predominant electricity sources by 2030, the share of renewable energy is projected to rise significantly by 2050, driven by stringent CO₂ reduction targets. In particular, natural gas combined with carbon capture and storage (NG-CCS) is expected to play a crucial role in balancing reliable energy supply with reduced

emissions, complemented by biomass-CCS technologies. This work also shows that under accelerated technology development paths, the power sector naturally develops toward a larger participation of renewables, thereby reducing the cost premium of achieving CO₂ targets to just 1-1.8% over scenarios without CO₂ policies. Furthermore, our analysis shows that the North Carolina energy system is resilient against hurricane conditions, having its capacity expansion solutions only mildly affected by the risk of hurricanes leading to a maximum additional cost increase of 0.8% on average (from 2023 to 2050) under the most challenging policy conditions, compared to scenarios that do not consider the risk of hurricanes.

5.1 Introduction

The power sector represents a substantial component of the global economy, with annual investments reaching USD 1.2 trillion worldwide, of which more than 80% have been directed towards renewable energy and grid investments (EIA 2023d). When looking at all net zero CO₂ targets and climate commitments around the world, EIA estimates this sector may increase by more than 40% by 2030 (EIA 2023d). Given its rapid growth and substantial impact on people's lives, multiple researchers are constantly looking at ways to plan the development of power systems such that risk and costs can be minimized, considering future uncertainties on renewable energy prices (Gagnon et al. 2023), fuel prices (EIA 2023a), government policies (C2ES 2024; DOE 2023), and extreme weather conditions (Bennett et al. 2021; Dyreson et al. 2022).

In this context, capacity expansion models have become a widely used tool by major research institutions (EIA 2022c; NREL 2020) and utilities (Duke Energy 2022b; FPL 2024). These models simulate the evolution of power systems by creating a representation of the existing system components and future technology options with their corresponding costs, efficiencies, and technical limitations, together with electricity demand forecasts and government targets. Capacity expansion models for the power sector are usually represented as linear optimization problems where the objective function is to find the optimal combination of future energy resources that meet energy demand with minimum cost (Hunter et al. 2013; NREL 2020).

Due to the uncertainties in estimating future economic variables and government policies, the use of sensitivity and scenario analysis has become a common practice in many capacity expansion studies (Gagnon et al. 2023; EIA 2023a). In this case, the most

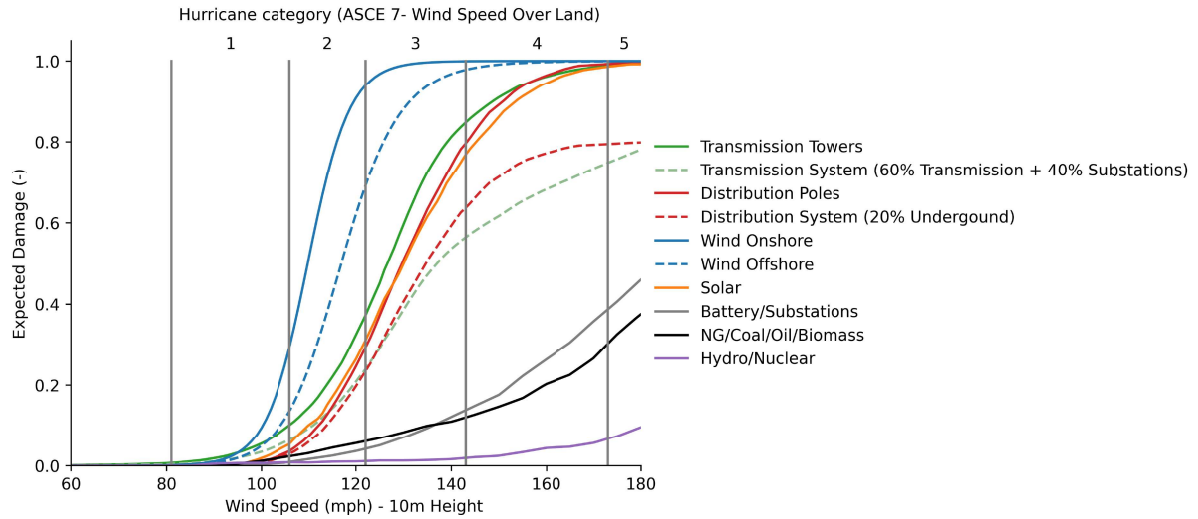


Figure 5.4: Summary of Fragility Curves Used in the Capacity Expansion Model for the Computation of Damage to the Power System Infrastructure. (Wind Speed: 3s Peak Gust at 10m Height).

Regarding hurricane wind speed statistics in the region of analysis, this work utilized data from the FEMA-HAZUS Hurricane model (FEMA 2022a). This model simulates synthetic hurricane conditions over 100,000 years on the US East Coast and has been thoroughly validated with historical observations (FEMA 2022a), having also been used by other researchers on modeling the impact of extreme hurricane conditions on the power system infrastructure (Schumann and Chini 2023; Mensah and Dueñas-Orsorio 2016). Figures 5.5 and 5.6 illustrate the 100-year and 1000-year hurricane statistics from the FEMA-HAZUS model.

Although the location and capacity of existing generating units are known from EIA reports (EIA 2022a) (see Figure 5.1), which allows the assignment of individual hurricane wind speed statistics and corresponding damages (technology, location, and hurricane dependent), the same cannot be said regarding the new energy generating units. The location of new deployments is a decision of the optimization model, and due to limitations in computational complexity, only a limited number of regions is often modeled in capacity expansion studies, as discussed in Section 5.2.1. This work divides NC into three regions, as shown in Figure 5.1. Based on the 1000-year hurricane wind data from HAZUS (Figure 5.6), region R1 is defined as any place in NC with wind speeds above 130mph, region R2 is defined by wind speeds between 111-130mph, and region R3 is defined by wind speeds below 111mph.

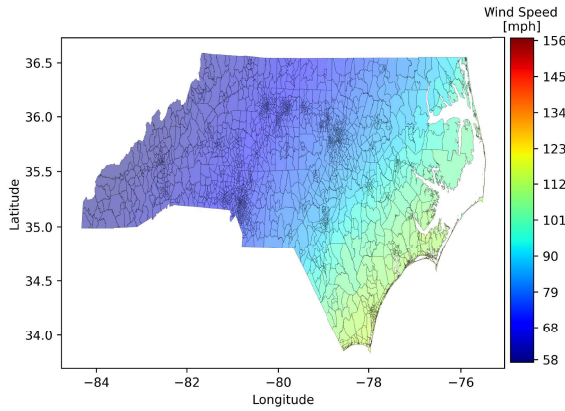


Figure 5.5: One Every 100 Years Extreme Wind Speeds From HAZUS (peak gust at 10m height).

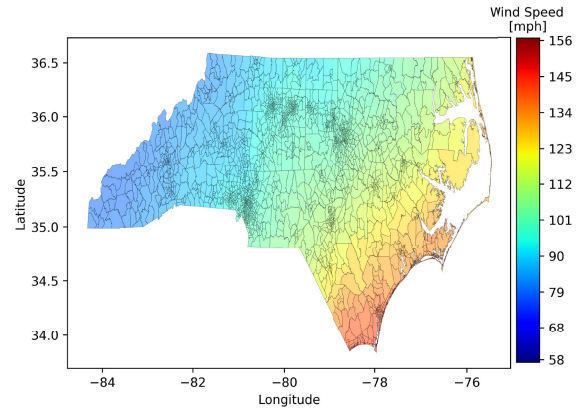


Figure 5.6: One Every 1000 Years Extreme Wind Speeds From HAZUS (peak gust at 10m height).

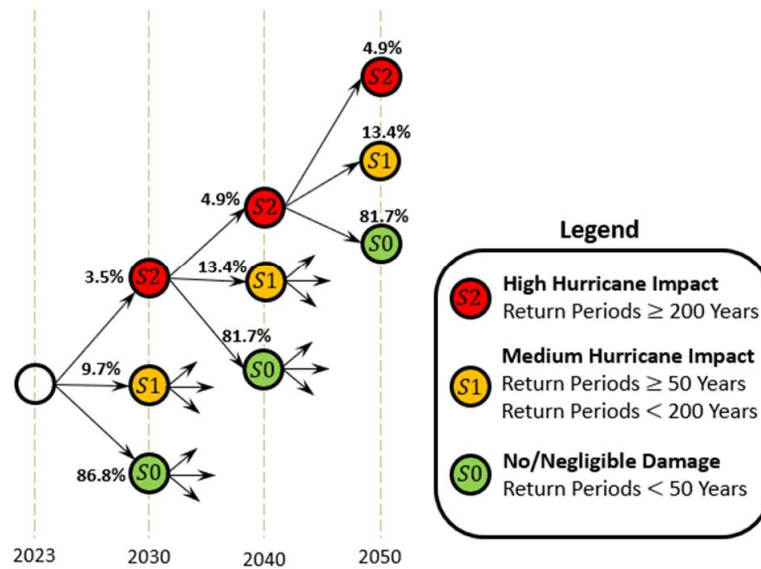


Figure 5.7: Scenario Tree Considered in the Modeling of the North Carolina Power Sector (Total of 27 Scenarios). The Hurricanes Realised in 2030 Encompass a 7-Year Window (2023-2030), and the Hurricanes Realised in 2040 and 2050 Encompass 10-Year Windows (2030-2040 and 2040-2050).

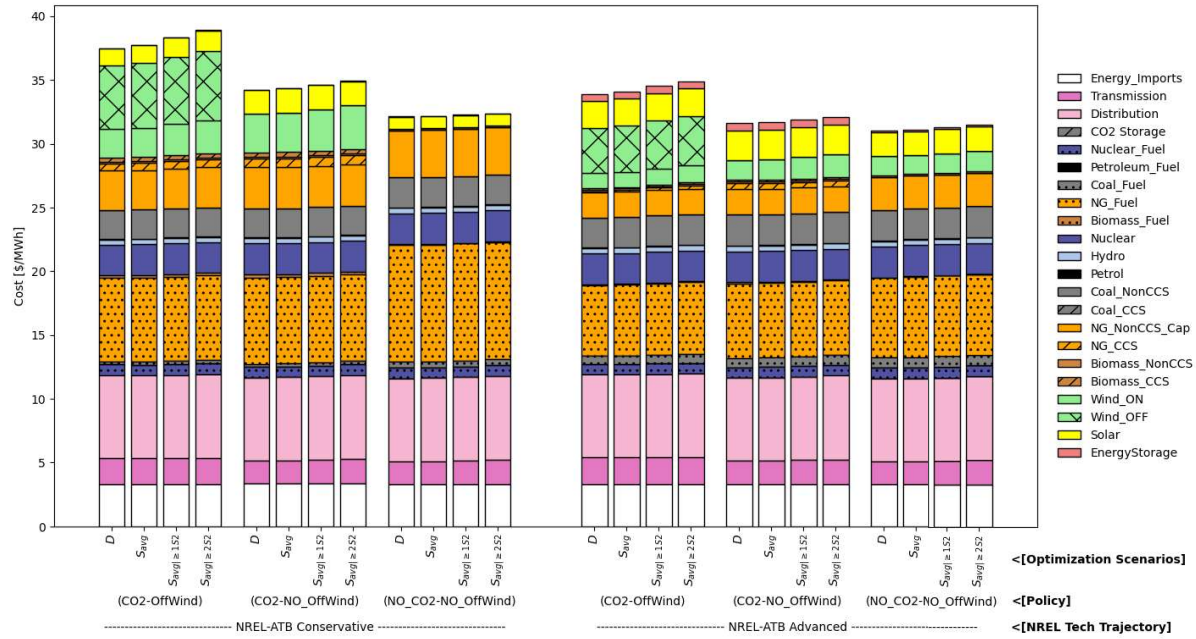


Figure 5.13: Total Cost Of Each Technology Group from 2023 to 2050 (Discounted to 2022 USD) Divided by Total Electricity Demand for the Entire NC System. D : Deterministic Model Simulation Without The Risk of Hurricane; S_{avg} : Expected Solution of the Stochastic Model Simulation With This Risk of Hurricane; $S_{avg|\geq 1S2}$: S_{avg} Conditioned that At Least one High Intense Hurricane (S2) Happens; $S_{avg|\geq 2S2}$: S_{avg} Conditioned that At Least Two High Intense Hurricane (S2) Happens. All simulations were done using Reference EIA Fuel Prices.

Finally, Appendix D.17 provides additional analysis related to the effect of increased hurricane intensity on the capacity expansion solution (+10% and +15% wind speed increase), arriving at similar trends from the ones presented in this section, although with slight increases in cost and participation of NG-CCS technology.

5.4 Conclusion

This work investigates the long-term planning of the North Carolina Energy system using capacity expansion models to evaluate the impact of different government policies, fuel prices, and technology evolution trajectories on total electricity costs and the participation of each generation technology in the total energy mix from 2023 to 2050.

Our findings indicate that by 2030, traditional energy sources such as nuclear, natural gas (NG), and coal will remain predominant in electricity generation, with coal increasing

its share, particularly when fuel prices rise and in scenarios without CO₂ targets, often taking space from NG generation. Nonetheless, in all scenarios with CO₂ targets by 2030 (70% CO₂ reduction from 2005), the model finds deploying solar and land-based wind more economical than using carbon capture technologies (CCS) to balance CO₂ emissions.

In scenarios with net zero CO₂ targets by 2050, renewables (solar, wind, and biomass) will become the major providers of electricity. However, NG with carbon capture (NG-CCS) also plays a significant role in many of these scenarios, generating a substantial fraction of the total electricity with minor atmospheric CO₂ emissions, which are compensated by CO₂ captured by biomass CCS technologies. Furthermore, even without the enforcement of CO₂ targets, our models show that renewables could still play a significant role if technologies advance along moderate or advanced trajectories (NREL-ATB paths).

By 2030, our models do not indicate the need for large energy storage requirements, but this scenario changes significantly on the path to 2050 when storage becomes a critical component of the system operation. Offshore wind is not found to be a competitive technology option in any of the scenarios simulated, only being considered when offshore energy targets are enforced (Ex. Order 218 (NC-GOV 2021a)).

In terms of cost, it is possible to notice a critical influence of technology development trajectory in reducing the impact of fuel price fluctuations and mitigating cost increases from the enforcement of CO₂ targets. However, enforcing offshore wind energy targets has shown to be one of the most significant factors in increasing the system's operational costs.

This work also explores the use of stochastic optimization in modeling the influence of low-probability, high-impact hurricane events on the results of capacity expansion models. The study found that the planning of the North Carolina power sector was not substantially affected by considering or not the risk of hurricanes if offshore energy technologies are not part of the system portfolio. In this case, loss of activity from offshore infrastructure is substituted by NG, solar, and on-shore wind generation. Nonetheless, even for scenarios with offshore wind, the difference in the total system cost for simulations with and without hurricanes is not larger than 0.8% on average (expected value of all hurricane scenarios), or around 1.5 Billion 2022 USD.

In terms of future work, the authors suggest investigating the impacts of the Inflation Reduction Act (US-Congress 2022) on the model solutions, as the substantial tax benefits from this government policy could significantly affect our findings in both analyses with and without hurricanes.

In terms of the stochastic model, the authors suggest the investigation of decomposition strategies in accelerating the simulation of the stochastic model, allowing the consideration